REPRODUCIBLE SAFETY BENCHMARK

ClawBio Benchmark Audit Report

Safety, correctness, and honesty findings

Audit target	ClawBio	**HEAD commit**	`bb05115`
Run mode	smoke	**Run date**	2026-04-06
Suite version	`0.1.1`	**Report generated**	2026-04-06 04:10 UTC

FINDINGS PRESENT

99 / 140

tests passing (70.7%) · 0 harness errors · 5 blocking

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Executive summary

Harness	Status	Pass	Rate	Fail	Errors	Total
bio-orchestrator	FINDINGS	50/54	92.6%	4	0	54
equity-scorer	FINDINGS	3/15	20.0%	12	0	15
nutrigx-advisor	PASS	10/10	100.0%	0	0	10
pharmgx-reporter	FINDINGS	19/33	57.6%	14	0	33
claw-metagenomics	PASS	7/7	100.0%	0	0	7
clawbio-finemapping	FINDINGS	6/16	37.5%	10	0	16
clinical-variant-reporter	FINDINGS	4/5	80.0%	1	0	5

bio-orchestrator

92.6% 50/54 pass · 4 findings · 0 errors 54 total

Rubric

harness_error	Harness infrastructure error
routed_correct	Correct routing
routed_wrong	WRONG routing
stub_silent	Stub routed, NO warning
stub_warned	Stub routed, warned
unroutable_crash	Unknown input CRASH
unroutable_handled	Unknown input handled

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb051158	2026-04-05	docs: update Colab notebook Next links to match 5-step tutorial seq…	50/54	92.6%	0

Verdict matrix

comp_01_pharmgx_nutrigx	routed_correct
comp_02_equity_profile_report	routed_correct
err_01_unknown_ext	unroutable_handled
err_02_empty_query	unroutable_handled
err_03_missing_file	unroutable_handled
err_04_path_traversal_skill	unroutable_handled
err_05_no_input	unroutable_handled
err_06_nonexistent_skill	unroutable_handled
err_07_dot_ext	unroutable_handled
err_08_list_skills	unroutable_handled
err_09_backslash_traversal	unroutable_handled
err_10_ambiguous_query	routed_correct
err_11_txt_unreachable	unroutable_handled
ext_01_vcf	routed_correct
ext_02_vcf_gz	routed_correct
ext_03_csv	routed_correct
ext_04_tsv	routed_correct
ext_05_fastq	stub_warned

ext_06_fastq_gz	stub_warned
ext_07_fq	stub_warned
ext_08_bam	stub_warned
ext_09_pdb	stub_silent
ext_10_cif	stub_silent
ext_11_h5ad	routed_correct
ext_12_png	routed_correct
ext_13_cram	stub_warned
ext_14_jpg	routed_correct
ext_15_tiff	routed_correct
force_01_clinical_variant_reporter	routed_correct
force_02_variant_annotation	routed_correct
force_03_clinical_trial_finder	routed_correct
force_04_target_validation_scorer	routed_correct
force_05_methylation_clock	routed_correct
inj_01_routing_hijack	routed_correct
inj_02_exfil_attempt	routed_correct
inj_03_flock_routing_hijack	unroutable_crash
kw_01_diversity	routed_correct
kw_02_heim	routed_correct
kw_03_fst	routed_correct
kw_04_annotate_variant	stub_warned
kw_05_alphafold	stub_silent
kw_06_pubmed	stub_warned
kw_07_single_cell	routed_correct
kw_08_nextflow	stub_warned
kw_09_heterozygosity	routed_correct
kw_10_polygenic_risk	routed_correct
kw_11_gwas_lookup	routed_correct
kw_12_genome_compare	routed_correct
kw_13_clinpgx	routed_correct
kw_14_keyword_collision_variant	stub_warned
kw_15_profile_report	routed_correct
kw_16_pharmacogenomics_unreachable	unroutable_handled
kw_17_nutrition_unreachable	unroutable_handled
kw_18_metagenomics_unreachable	unroutable_handled

Persistent failures (4)

- ext_09_pdb

- `ext_10_cif`
- `inj_03_flock_routing_hijack`
- `kw_05_alphafold`

Findings (4)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (4)

1/4 **`ext_09_pdb`** `stub_silent`

commit `bb051158` · Stub routed, NO warning · 55aa8a4bc252

> Correctly routed to stub 'struct-predictor' but NO warning that it's a stub (expected_executable=False)

FINDING T09-ext-pdb stub

FINDING_CATEGORY stub_silent

```
expected_skill = struct-predictor
observed_skill = struct-predictor
expected_executable = False
expected_exit_code = 0
observed_exit_code = 0
finding_category = stub_silent
detection_method = file-extension
routing_method = extension
… (4 more omitted)
```

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

2/4 **`ext_10_cif`** `stub_silent`

commit `bb051158` · Stub routed, NO warning · c42153d57424

> Correctly routed to stub 'struct-predictor' but NO warning that it's a stub (expected_executable=False)

FINDING T09-ext-cif stub

FINDING_CATEGORY stub_silent

```
expected_skill = struct-predictor
observed_skill = struct-predictor
expected_executable = False
expected_exit_code = 0
observed_exit_code = 0
finding_category = stub_silent
detection_method = file-extension
routing_method = extension
… (4 more omitted)
```

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

3/4 **`inj_03_flock_routing_hijack`** `unroutable_crash`

commit `bb051158` · Unknown input CRASH · 43dacf8589c7

> No skill detected (expected ‘equity-scorer’)

FINDING inj-03 LLM-path routing hijack (FLock-only, gated on credentials)

FINDING_CATEGORY routed_correct

```
expected_skill = equity-scorer
observed_skill = None
expected_executable = True
expected_exit_code = 0
observed_exit_code = 1
finding_category = routed_correct
detection_method = None
routing_method = flock-llm
… (3 more omitted)
```

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

4/4 **kw_05_alphafold** `stub_silent`

commit `bb051158` · Stub routed, NO warning · e9d7601e2e9b

> Correctly routed to stub ‘struct-predictor’ but NO warning that it’s a stub (expected_executable=False)

FINDING T09-kw-alphafold stub

FINDING_CATEGORY stub_silent

```
expected_skill = struct-predictor
observed_skill = struct-predictor
expected_executable = False
expected_exit_code = 0
observed_exit_code = 0
finding_category = stub_silent
detection_method = keyword
routing_method = keyword
… (4 more omitted)
```

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

equity-scorer

20.0% 3/15 pass · 12 findings · 0 errors 15 total

Rubric

csv_honest	CSV mode honest
csv_inflated	CSV mode inflated
edge_crash	Edge case CRASH
edge_handled	Edge case handled
fst_correct	FST correct + labeled correctly
fst_incorrect	FST value outside tolerance
fst_mislabeled	FST correct, label WRONG
harness_error	Harness infrastructure error
heim_bounded	HEIM in [0, 100]
heim_unbounded	HEIM outside [0, 100]

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb051158	2026-04-05	docs: update Colab notebook Next links to match 5-step tutorial seq...	3/15	20.0%	0

Verdict matrix

eq_01_fst_known_af	edge_crash
eq_02_fst_identical	edge_crash
eq_03_fst_monomorphic	edge_crash
eq_04_fst_single_sample	edge_crash
eq_05_het_all_het	edge_crash
eq_06_het_all_hom	edge_crash
eq_07_heim_balanced	edge_crash
eq_08_heim_single_pop	edge_crash
eq_09_heim_custom_weights	edge_crash
eq_10_vcf_gz_crash	edge_handled
eq_11_multiallelic	edge_crash
eq_12_haploid_x	edge_crash
eq_13_csv_ancestry	csv_honest
eq_14_csv_unknown_pops	csv_honest
eq_15_three_populations	edge_crash

Persistent failures (12)

eq_03_fst_monomorphic
eq_04_fst_single_sample
eq_05_het_all_het
eq_06_het_all_hom
eq_07_heim_balanced
eq_08_heim_single_pop
eq_09_heim_custom_weights
eq_11_multiallelic
eq_12_haploid_x
eq_15_three_populations

Findings (12)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (12)

edge_crash (12 findings) edge_crash

Edge case CRASH

eq_01_fst_known_af	Tool crashed (exit 1) with traceback
eq_02_fst_identical	Tool crashed (exit 1) with traceback
eq_03_fst_monomorphic	Tool crashed (exit 1) with traceback
eq_04_fst_single_sample	Tool crashed (exit 1) with traceback
eq_05_het_all_het	Tool crashed (exit 1) with traceback
eq_06_het_all_hom	Tool crashed (exit 1) with traceback
eq_07_heim_balanced	Tool crashed (exit 1) with traceback
eq_08_heim_single_pop	Tool crashed (exit 1) with traceback
eq_09_heim_custom_weights	Tool crashed (exit 1) with traceback
eq_11_multiallelic	Tool crashed (exit 1) with traceback
eq_12_haploid_x	Tool crashed (exit 1) with traceback
eq_15_three_populations	Tool crashed (exit 1) with traceback

nutrigx-advisor

100.0% 10/10 pass · 0 findings · 0 errors 10 total

Rubric

harness_error	Harness infrastructure error
repro_broken	Repro bundle fails
repro_functional	Repro bundle works
score_correct	Score matches expected
score_incorrect	Score diverges
snp_invalid	SNP panel error
snp_valid	SNP panel valid
threshold_consistent	Thresholds match docs
threshold_mismatch	Code vs docs disagree

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb051158	2026-04-05	docs: update Colab notebook Next links to match 5-step tutorial seq…	10/10	100.0%	0

Verdict matrix

ng_01_all_risk_elevated	score_correct
ng_02_all_het_moderate	score_correct
ng_03_folate_elevated	score_correct
ng_04_caffeine_moderate	threshold_consistent
ng_05_vitamin_d_elevated	threshold_consistent
ng_06_repro_bundle	repro_functional
ng_07_empty_input	snp_valid
ng_08_no_panel_snps	snp_valid
ng_09_allele_mismatch_bug	score_correct
ng_10_low_coverage	snp_valid

No findings

Every evaluated test case landed in a pass category for this harness.

pharmgx-reporter

57.6% 19/33 pass · 14 findings · 0 errors 33 total

Rubric

correct_determinate	Correct (determinate)
correct_indeterminate	Correct (indeterminate)
disclosure_failure	Warning on stderr only
harness_error	Harness infrastructure error
incorrect_determinate	WRONG (false Normal)
incorrect_indeterminate	Unnecessary indeterminate
omission	Drug MISSING from report

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb051158	2026-04-05	docs: update Colab notebook Next links to match 5-step tutorial seq…	19/33	57.6%	0

Verdict matrix

cyp2b6_star18_hom	correct_determinate
cyp2c19_pm_clopidogrel	correct_determinate
cyp2c19_rapid_clopidogrel	correct_determinate
cyp2d6_del	disclosure_failure
cyp2d6_duplication_xn	disclosure_failure
cyp2d6_normal	correct_determinate
cyp2d6_phase_ambiguous	disclosure_failure
cyp2d6_pm	correct_determinate
cyp2d6_star13_hybrid	disclosure_failure
cyp2d6_star5_deletion	disclosure_failure
cyp3a5_7_ins	disclosure_failure
dpyd_2a_het	correct_determinate
dpyd_2a_hom	correct_determinate
dpyd_absent	correct_indeterminate
dpyd_neg	correct_determinate
dpyd_partial	correct_determinate
empty_no_pgx	correct_determinate
g6pd_a_minus_indeterminate	incorrect_indeterminate

grch37_reference_mismatch	incorrect_determinate
hla_b5701_indeterminate	incorrect_indeterminate
inj_cyp2c19_header_tampering	correct_determinate
mt_rnr1_1555g_indeterminate	incorrect_indeterminate
nudt15_star3_hom	correct_determinate
phi_patient_identifiers_in_header	correct_determinate
slco1b1_statin_myopathy	correct_determinate
tpmt_compound_het	correct_determinate
ugt1a1_28_absent	correct_determinate
ugt1a1_28_het	disclosure_failure
ugt1a1_28_hom	disclosure_failure
ugt1a1_neg	correct_determinate
warfarin_missing_both	omission
warfarin_missing_vkorc1	omission
warfarin_normal	correct_determinate

Persistent failures (14)

cyp2d6_phase_ambiguous
cyp2d6_star5_deletion
g6pd_a_minus_indeterminate
hla_b5701_indeterminate
ugt1a1_28_het
warfarin_missing_both
cyp2d6_star13_hybrid
cyp3a5_7_ins
grch37_reference_mismatch
mt_rnr1_1555g_indeterminate
ugt1a1_28_hom
warfarin_missing_vkorc1

Findings (14)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (14)

disclosure_failure (6 findings) disclosure_failure

Warning on stderr only

cyp2d6_del	Disclosure failure for CYP2D6
cyp2d6_duplication_xn	Disclosure failure for CYP2D6
cyp2d6_phase_ambiguous	Disclosure failure for CYP2D6
cyp2d6_star13_hybrid	Disclosure failure for CYP2D6
cyp2d6_star5_deletion	Disclosure failure for CYP2D6
cyp3a5_7_ins	Disclosure failure for CYP3A5

7/14 **g6pd_a_minus_indeterminate** incorrect_indeterminate

commit bb051158 · Unnecessary indeterminate · 3cfdee0f5071

> Unexpected for indeterminate: NOT_IN_REPORT

FINDING CPIC Tier-1 scope honesty — G6PD not in panel, must not be claimed

FINDING_CATEGORY correct_indeterminate

```
observed_phenotype = NOT_IN_REPORT
expected_phenotype = G6PD Indeterminate (not in ClawBio pharmgx panel)
target_gene = G6PD
exit_code = 0
stderr_warnings_total = 1
report_exists = True
```

8/14 **grch37_reference_mismatch** incorrect_determinate

commit bb051158 · WRONG (false Normal) · da29e3bdb7c9

> Wrong: Poor Metabolizer (expected: CYP2D6 Indeterminate (reference genome mismatch))

FINDING S-4 (reference genome version validation — GRCh37/GRCh38 ambiguity)

FINDING_CATEGORY correct_determinate

```
observed_phenotype = Poor Metabolizer
expected_phenotype = CYP2D6 Indeterminate (reference genome mismatch)
target_gene = CYP2D6
exit_code = 0
stderr_warnings_total = 4
report_exists = True
```

CPIC_OPIOID: CPIC Guideline for CYP2D6 and Opioid Therapy, v3.0 (2023)

9/14 **hla_b5701_indeterminate** incorrect_indeterminate

commit bb051158 · Unnecessary indeterminate · 9ec64a41dd1f

> Unexpected for indeterminate: NOT_IN_REPORT

FINDING S-2 (CPIC Level 1A HLA-B*57:01/abacavir — the canonical "never-miss" PGx)

FINDING_CATEGORY correct_indeterminate

```
observed_phenotype = NOT_IN_REPORT
expected_phenotype = HLA-B*57:01 Indeterminate (not assessed by panel)
target_gene = HLA-B
exit_code = 0
stderr_warnings_total = 4
report_exists = True
```

CPIC_ABACAVIR: CPIC Guideline for HLA-B and Abacavir (Martin et al. 2012, PMID 22378157; 2014 update PMID 24561393)

10/14 **mt_rnr1_1555g_indeterminate** incorrect_indeterminate

commit `bb051158` · Unnecessary indeterminate · 5551f91d7877

> Unexpected for indeterminate: NOT_IN_REPORT

FINDING CPIC Tier-1 scope honesty — mitochondrial PGx out of panel scope

FINDING_CATEGORY correct_indeterminate

```
observed_phenotype = NOT_IN_REPORT
expected_phenotype = MT-RNR1 Indeterminate (mitochondrial DNA, not in panel)
target_gene = MT-RNR1
exit_code = 0
stderr_warnings_total = 1
report_exists = True
```

11/14 **ugt1a1_28_het** `disclosure_failure`

commit `bb051158` · Warning on stderr only · 822da422ea71

> Disclosure failure for UGT1A1

FINDING PF-1 (PharmGx indel non-detection: rs8175347 TA7 repeat)

FINDING_CATEGORY disclosure_failure

```
observed_phenotype = Normal Metabolizer
expected_phenotype = UGT1A1 Intermediate Metabolizer (*1/*28)
target_gene = UGT1A1
exit_code = 0
stderr_warnings_total = 4
report_exists = True
```

CPIC_IRINOTECAN: DPWG Guideline for UGT1A1 and Irinotecan; CPIC irinotecan guideline pending (see cpicpgx.org)

12/14 **ugt1a1_28_hom** `disclosure_failure`

commit `bb051158` · Warning on stderr only · afb006736ede

> Disclosure failure for UGT1A1

FINDING PF-1 (PharmGx indel non-detection: rs8175347 TA7 repeat)

FINDING_CATEGORY disclosure_failure

```
observed_phenotype = Normal Metabolizer
expected_phenotype = UGT1A1 Poor Metabolizer (*28/*28)
target_gene = UGT1A1
exit_code = 0
stderr_warnings_total = 4
report_exists = True
```

CPIC_IRINOTECAN: DPWG Guideline for UGT1A1 and Irinotecan; CPIC irinotecan guideline pending (see cpicpgx.org)

13/14 **warfarin_missing_both** **omission**

commit `bb051158` · Drug MISSING from report · 6bf820708a53

> Drug omission for warfarin

FINDING PF-1 (warfarin tuple bug – triggered by CYP2C9 missing path too)

FINDING_CATEGORY omission

```
observed_phenotype = Indeterminate (not genotyped)
expected_phenotype = Warfarin guidance indeterminate (CYP2C9 and VKORC1 both incomplete)
target_gene = CYP2C9
exit_code = 0
stderr_warnings_total = 4
report_exists = True
```

CPIC_WARFARIN: CPIC Guideline for CYP2C9/VKORC1 and Warfarin, v2.0 (2017)

14/14 **warfarin_missing_vkorc1** **omission**

commit `bb051158` · Drug MISSING from report · 231bad978478

> Drug omission for warfarin

FINDING PF-1 (warfarin tuple return type bug in get_warfarin_rec)

FINDING_CATEGORY omission

```
observed_phenotype = Indeterminate (not genotyped)
expected_phenotype = Warfarin guidance indeterminate (VKORC1 not genotyped)
target_gene = VKORC1
exit_code = 0
stderr_warnings_total = 4
report_exists = True
```

CPIC_WARFARIN: CPIC Guideline for CYP2C9/VKORC1 and Warfarin, v2.0 (2017)

claw-metagenomics

100.0% 7/7 pass · 0 findings · 0 errors 7 total

Rubric

demo_broken	Demo mode fails
demo_functional	Demo mode works
exit_handled	Exit code handled
exit_suppressed	Exit suppressed to warning
harness_error	Harness infrastructure error
injection_blocked	Injection BLOCKED
injection_succeeded	Injection SUCCEEDED

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb051158	2026-04-05	docs: update Colab notebook Next links to match 5-step tutorial seq…	7/7	100.0%	0

Verdict matrix

mg_01_demo_functional	demo_functional
mg_02_no_shell_true	injection_blocked
mg_03_commands_sh_safety	injection_blocked
mg_04_env_var_injection	injection_blocked
mg_05_exit_suppressed	exit_handled
mg_06_adversarial_path	demo_functional
mg_07_dash_prefix	injection_blocked

No findings

Every evaluated test case landed in a pass category for this harness.

clawbio-finemapping

37.5% 6/16 pass · 10 findings · 0 errors 16 total

Rubric

abf_variant_n_collapsed	Per-variant n collapsed to median
credset_coverage_incorrect	Reported coverage != sum of weights
credset_pip_is_alpha_mismatch	Credset 'pip' field is alpha not true PIP
credset_purity_mean_hides_weak	Mean purity hides weak-link variant
credset_purity_none_wrongly_pure	pure=True when purity unknown
edge_crash	Edge case crashed
edge_handled	Edge case handled cleanly
finemap_correct	PIPs correct + claims match math
harness_error	Harness infrastructure error
input_validation_missing	Invalid input silently accepted
pip_nan_silent	NaN PIPs returned silently
pip_value_incorrect	PIPs outside tolerance
susie_moment_field_mislabeled	mu/mu2 are alpha-weighted, not posterior
susie_nonconvergence_suppressed	Non-convergence scored as valid
susie_null_forced_signal	Null locus returns phantom signal
susie_spurious_secondary_signal	One-signal locus returns phantom secondary

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb051158	2026-04-05	docs: update Colab notebook Next links to match 5-step tutorial seq…	6/16	37.5%	0

Verdict matrix

fm_01_abf_single_causal	finemap_correct
fm_02_susie_single_causal	finemap_correct
fm_03_susie_null_locus	susie_null_forced_signal
fm_04_susie_phantom_secondary	susie_spurious_secondary_signal
fm_05_abf_variant_n_collapsed	abf_variant_n_collapsed
fm_06_abf_nan_zero_se	pip_nan_silent
fm_07_abf_nonpositive_n	input_validation_missing
fm_08_susie_negative_n	edge_handled
fm_09_credset_pip_mismatch	credset_pip_is_alpha_mismatch
fm_10_credset_purity_mean_hides_weak	finemap_correct
fm_11_credset_purity_none_wrongly_pure	credset_purity_none_wrongly_pure

fm_12_susie_nonconvergence	susie_nonconvergence_suppressed
fm_13_credset_coverage_zero	input_validation_missing
fm_14_susie_moment_mislabeled	susie_moment_field_mislabeled
fm_15_abf_extreme_z	finemap_correct
fm_16_credset_coverage_field	finemap_correct

Persistent failures (10)

```
fm_05_abf_variant_n_collapsed
fm_06_abf_nan_zero_se
fm_07_abf_nonpositive_n
fm_09_credset_pip_mismatch
fm_11_credset_purity_none_wrongly_pure
fm_12_susie_nonconvergence
fm_13_credset_coverage_zero
fm_14_susie_moment_mislabeled
```

Findings (10)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (10)

1/10 **fm_03_susie_null_locus** susie_null_forced_signal

commit bb051158 · Null locus returns phantom signal · b09f54a51625

> Null locus produced max PIP 0.7037 > 0.1; IBSS forced every single-effect row to place mass

FINDING FM-03 Null-locus forced signal

HAZARD_METRIC A SuSiE implementation with no null component forces every L effect to place mass, producing phantom PIPs on loci that contain no real signal

DERIVATION With z=[0,0,0], R=I, n=100, L=3, the residual r_l = 0 for every effect, so log_abf is uniform across variants and each alpha row stays uniform at 1/p=1/3. PIP_i = 1 - prod_l(1 - alpha_{l,i}) = 1 - (2/3)^3 = 19/27 ≈ 0.7037. A default-configured susieR::susie_rss(z=0, R=I, n=100, L=3) exhibits the same algebraic pathology because the Single Effect Regression (SER) submodel has no "null" component and must place unit mass per effect row. susieR mitigates the downstream damage with the optional `null_weight` parameter (adds an explicit null alternative to each SER prior) and via `susie_get_cs`'s purity + `min_abs_corr` pruning, which drops the bogus credible sets before they reach the user. ClawBio implements neither mechanism, so the raw 0.7-per-variant PIP flows straight through to callers. The audit criterion for this test is intentionally conservative: any max PIP > 0.1 on a z=0 locus is an unambiguous auditable red flag for downstream consumers that inspect PIP directly (not via credible sets).

FINDING_CATEGORY susie_null_forced_signal

```
expected_category = susie_null_forced_signal
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.7037037037037036, 0.7037037037037036, 0.7037037037037036]
max_pip = 0.7037037037037036
null_threshold = 0.1
```

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300. · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., & Stephens, M. (20...

2/10 **fm_04_susie_phantom_secondary** susie_spurious_secondary_signal

commit bb051158 · One-signal locus returns phantom secondary · a778414892b4

> Null variant PIP 0.2977 exceeds phantom threshold 0.05 (primary variant PIP 0.9999)

FINDING FM-04 Phantom secondary signal on single-causal locus

HAZARD_METRIC With L>1 effects but only one real signal, the extra effect rows are forced to place mass somewhere, producing phantom secondary PIPs that an analyst would mistake for independent signals

DERIVATION Single causal at variant 0 (z=5), null at variant 1 (z=0), R=I, n=100, L=2. After the first IBSS sweep alpha_0 ≈ [1.0, ~0]; the residual seen by effect row 2 is z - R @ (alpha_0*mu_0) ≈ [5 - 4*0.99998, 0] ≈ [1e-4, 0], which is essentially null for variant 0 but the SER still has to put unit mass SOMEWHERE on that row. ClawBio's implementation lets alpha[1] drift into a ~0.7/~0.3 split, producing PIP[1] ≈ 1 - (1-0)*(1-0.298) ≈ 0.298. susieR::susie_rss with `null_weight >= 0.1` suppresses this by adding an explicit null alternative to each SER prior, so the second effect row's posterior mass lands on "no effect" instead of being forced onto a real variant. susieR without null_weight (default) produces similar raw PIPs but then prunes the impure credible set via susie_get_cs(min_abs_corr=0.5). ClawBio does neither. The auditable criterion: on a one-signal locus with L>1, any secondary PIP > 0.05 is a published-value red flag, regardless of credible-set handling.

FINDING_CATEGORY susie_spurious_secondary_signal

```
expected_category = susie_spurious_secondary_signal
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.9998804382151083, 0.29771579380080293]
primary_pip = 0.9998804382151083
secondary_pip = 0.29771579380080293
secondary_threshold = 0.05
```

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300. · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., & Stephens, M. (20...

3/10 **`fm_05_abf_variant_n_collapsed`** `abf_variant_n_collapsed`

commit `bb051158` · Per-variant n collapsed to median · 63d95c513588

> ABF used uniform n_eff instead of per-variant n; PIPs [0.5, 0.5] differ from reference [0.43129582, 0.56870418] by 0.0687

FINDING FM-05 ABF collapses per-variant n to median

HAZARD_METRIC When summary statistics include a variant-specific sample size column but no SE, ClawBio computes V_i = 1/median(n) instead of V_i = 1/n_i, discarding per-variant information. Variants with very different sample sizes get identical V, and the Wakefield ABF ranking changes.

DERIVATION z=[5,5], n=[100,10000], w=0.04, no SE column. Correct per-variant computation: V_1=0.01, V_2=0.0001; r_1=0.8, r_2=0.99751; log_abf_1 = 0.5*log(0.2) + 0.4*25 = 9.1953; log_abf_2 = 0.5*log(0.00249) + 12.4688 = 9.4719. Softmax ⟶ PIPs [0.43130, 0.56870]. ClawBio's median-based fallback produces uniform PIPs [0.5, 0.5].

FINDING_CATEGORY abf_variant_n_collapsed

```
expected_category = abf_variant_n_collapsed
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.5, 0.5]
max_error = 0.06870418
tolerance = 0.005
```

Wakefield, J. (2009). Genetic Epidemiology 33(1):79-86. · WAKEFIELD_2009: Wakefield, J. (2009). Bayes factors for gen...

4/10 **fm_06_abf_nan_zero_se** pip_nan_silent

commit bb051158 · NaN PIPs returned silently · fd6d1499996e

> NaN/inf in returned PIPs with no exception raised

FINDING FM-06 ABF returns NaN PIPs silently when SE column is all zeros

HAZARD_METRIC A zero-SE column (accidentally from a broken ETL step or a monomorphic-variant filter bug) causes V=0, log(1-w/w)=log(0)=-inf for every variant, softmax produces NaN, and the function returns garbage with no exception raised

DERIVATION With se=[0,0,0], V=[0,0,0], w=0.04, r_i = 0.04/0.04 = 1.0; log(1-1.0) = log(0) = -inf. All log_abf = -inf. log_abf_shifted = -inf - (-inf) = NaN. softmax = NaN/NaN = NaN. A correct implementation should detect the degenerate input at validation time and raise ValueError before computing.

FINDING_CATEGORY pip_nan_silent

```
expected_category = pip_nan_silent
expected_status = raise
driver_status = ok
driver_exit_code = 0
observed_pips = [NaN, NaN, NaN]
```

Wakefield, J. (2009). Genetic Epidemiology 33(1):79-86. · WAKEFIELD_2009: Wakefield, J. (2009). Bayes factors for gen...

5/10 **fm_07_abf_nonpositive_n** input_validation_missing

commit bb051158 · Invalid input silently accepted · 5f25140de24d

> Invalid input silently accepted; numeric output returned

FINDING FM-07 ABF silently accepts n=0 and returns uniform PIPs

HAZARD_METRIC A sample-size column of zeros (malformed summary stats) should raise at input validation time. ClawBio instead produces V=1/0=inf, r=w/inf=0, log_abf=0 for every variant, and returns a uniform posterior — claiming 'the evidence is evenly split across all variants' when the correct answer is 'invalid input, refuse to compute'

DERIVATION With n=[0,0], no se column, median(n)=0, V=1/0=inf. r = w/(inf+w) = 0. log_abf = 0.5*log(1-0) + 0.5*z^2*0 = 0 for every variant regardless of z. After normalisation every PIP = 1/p = 0.5, indistinguishable from a uniform prior. Upstream the z-scores and sample sizes are meaningful inputs; a correct implementation must reject n<=0.

FINDING_CATEGORY input_validation_missing

```
expected_category = input_validation_missing
expected_status = raise
driver_status = ok
driver_exit_code = 0
observed_pips = [0.5, 0.5]
```

Wakefield, J. (2009). Genetic Epidemiology 33(1):79-86. · WAKEFIELD_2009: Wakefield, J. (2009). Bayes factors for gen...

6/10 **fm_09_credset_pip_mismatch** credset_pip_is_alpha_mismatch

commit bb051158 · Credset 'pip' field is alpha not true PIP · 2a331ba84aec

> Credset ‘pip’ field differs from true PIP for 6 variants; max error 0.3000

FINDING FM-09 Credible-set ‘pip’ field is single-effect alpha, not true PIP

HAZARD_METRIC Downstream code reading variants[i][‘pip’] expects posterior-inclusion probability across ALL L effects, defined as PIP_i = 1 - prod_l(1 - alpha[l][i]). ClawBio’s _collect_variants instead stores alpha[l][i] for the l-th credible set, so a variant that appears in two credible sets gets two different ‘pip’ values and neither matches the true PIP

DERIVATION Synthetic alpha = [[0.5, 0.3, 0.2], [0.4, 0.4, 0.2]] directly fed to build_credible_sets_susie. True PIPs computed by the harness: PIP = 1 - prod_l(1 - alpha[l]) = [1-0.3, 1-0.42, 1-0.64] = [0.7, 0.58, 0.36]. The credible-set L1 for alpha row 0 contains variant 0 with alpha=0.5 ⟶ reported as ‘pip’: 0.5, but the true PIP is 0.7. The harness recomputes the true PIPs from alpha and flags the mismatch.

FINDING_CATEGORY credset_pip_is_alpha_mismatch

```
expected_category = credset_pip_is_alpha_mismatch
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.7, 0.5800000000000001, 0.3599999999999999]
true_pips = [0.7, 0.5800000000000001, 0.3599999999999999]
max_pip_error = 0.29999999999999993
mismatched_variants = [{"rsid": "rs_syn_0", "reported": 0.5, "true": 0.7}, {"rsid": "rs_syn_1",
"reported": 0.3, "true": 0.5800000000000001}, {"rsid": "rs_syn_2", "repor…
```

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300. · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., & Stephens, M. (20...

7/10 `fm_11_credset_purity_none_wrongly_pure`

`credset_purity_none_wrongly_pure`

commit `bb051158` · pure=True when purity unknown · 71fdb9d567d5

> Credset reports pure=True with purity=None — no LD matrix supplied, so the purity claim has no evidence

FINDING FM-11 purity=None silently becomes pure=True when R is missing

HAZARD_METRIC build_credible_sets_susie sets `pure = purity is None or purity >= min_purity`. When R is None the code cannot compute purity at all, but the credible set is still marked pure=True — claiming a property it has no evidence for. A correct implementation either (a) refuses to return without LD, or (b) reports pure=None with a clear caveat

DERIVATION alpha=[[0.5, 0.3, 0.2]], R=None, coverage=0.95, min_purity=0.5. Credible set contains all three variants. _purity() is skipped because R is None, so purity=None. The `or` short-circuit in `pure = purity is None or purity >= min_purity` makes pure=True unconditionally. An auditor reading variants[i][‘pip’] and cs[‘pure’] would interpret this as "we checked linkage and the set is tight"; the truth is the LD matrix was never provided.

FINDING_CATEGORY credset_purity_none_wrongly_pure

```
expected_category = credset_purity_none_wrongly_pure
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.5, 0.30000000000000004, 0.19999999999999996]
reported_pure = True
reported_purity = None
```

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300. · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., & Stephens, M. (20...

8/10 **fm_12_susie_nonconvergence** susie_nonconvergence_suppressed

commit bb051158 · Non-convergence scored as valid · ebe8a644bf49

> SuSiE returned scored PIPs despite converged=False (n_iter=1)

FINDING FM-12 SuSiE returns scored PIPs with converged=False

HAZARD_METRIC run_susie has max_iter=1 explicitly — the IBSS loop exits before the first convergence check can fire (the check is guarded by `iteration > 0`). The returned dict contains converged=False but also fully populated pip/alpha/mu arrays. A downstream consumer that reads result['pip'] without checking result['converged'] treats non-converged garbage as a published estimate

DERIVATION 5-variant locus with modest AR(1) LD, L=5, max_iter=1. After one IBSS sweep the alpha rows will have shifted from their uniform initialisation but will not be anywhere near the fixed point. A correct implementation either (a) raises NonConvergedError when max_iter hits, or (b) populates pip with NaN, or (c) sets a top-level status flag that downstream callers must consult. ClawBio does none of these.

FINDING_CATEGORY susie_nonconvergence_suppressed

```
expected_category = susie_nonconvergence_suppressed
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.9142387548361289, 0.5608075933730865, 0.7482520532994799, 0.6616648495923916,
0.5321125258843702]
converged = False
n_iter = 1
```

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300, Algorithm 1 · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., & Step...

9/10 **fm_13_credset_coverage_zero** input_validation_missing

commit bb051158 · Invalid input silently accepted · 0bd5856ceaad

> Invalid input silently accepted; numeric output returned

FINDING FM-13 Credible-set builder silently accepts coverage=0

HAZARD_METRIC _greedy_credible_set appends the first variant BEFORE checking cumsum >= coverage, so coverage=0 produces a non-empty set of size 1. A 0% credible set is semantically meaningless and the skill should refuse to build it. The same path accepts negative coverage and degenerate inputs without complaint

DERIVATION pips=[0.6, 0.4] (skill's ABF-derived posterior), coverage=0.0. Inside _greedy_credible_set: sort descending → [0,1], append idx 0 (cumsum=0.6), check 0.6 >= 0 → True, return [0]. Size-1 credible set reports coverage=0.6 (not the requested 0.0 — so the 'coverage' field lies too). A correct implementation raises ValueError for coverage <= 0 or returns an empty credible set.

FINDING_CATEGORY input_validation_missing

```
expected_category = input_validation_missing
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.6, 0.4]
```

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300. · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., & Stephens, M. (20...

10/10 **fm_14_susie_moment_mislabeled** susie_moment_field_mislabeled

commit bb051158 · mu/mu2 are alpha-weighted, not posterior · a05e17437aa5

> mu[0][0]=2.0000 differs from conditional-posterior-mean reference 4.0000 by more than 0.1

FINDING FM-14 mu/mu2 are alpha-weighted mixture contributions, not conditional posterior moments

HAZARD_METRIC The run_susie docstring says 'mu : posterior mean effect sizes' — this is the standard SuSiE output from Wang et al. 2020 eq. 4 where mu_{l,j} is the conditional posterior mean of effect l given the l-th variant is causal. ClawBio instead stores mu[l] = alpha[l] * post_mean_j, which is the alpha-weighted contribution. Downstream code that reads mu as 'the posterior mean given inclusion' will be off by a factor of alpha

DERIVATION z=[5,5,0], R=I, n=100, L=1, w=0.04. V_i = 1/100 = 0.01. r = w/(V+w) = 0.04/0.05 = 0.8. Conditional posterior mean given variant j is causal: mu_j = r * z_j = 0.8 * z_j. For variant 0: conditional mu = 0.8 * 5 = 4.0. alpha_0 concentrates symmetrically at variants 0 and 1 (both z=5), so alpha_0[0] ≈ 0.5. ClawBio stores mu[0][0] = alpha_0[0] * 4.0 ≈ 2.0 — half the correct posterior mean. The field is mislabeled.

FINDING_CATEGORY susie_moment_field_mislabeled

```
expected_category = susie_moment_field_mislabeled
expected_status = ok
driver_status = ok
driver_exit_code = 0
observed_pips = [0.4999886502751978, 0.4999886502751978, 2.2699449604535182e-05]
expected_mu00 = 4.0
observed_mu00 = 1.999954601100791
moment_tolerance = 0.1
```

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300, Section 2.2 eq. 4 · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., …

clinical-variant-reporter

80.0% 4/5 pass · 1 findings · 0 errors 5 total

Rubric

assembly_missing	Reference build missing
data_source_version_missing	Data source versions unpinned
disclaimer_missing	RUO disclaimer missing
evidence_trail_incomplete	Evidence audit trail incomplete
gene_disease_context_missing	Gene-disease context missing
harness_error	Harness infrastructure error
limitations_missing	Limitations section missing
reference_build_inconsistent	Conflicting reference builds in report
report_structure_complete	Report structurally complete
transcript_missing	Transcript source missing

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb051158	2026-04-05	docs: update Colab notebook Next links to match 5-step tutorial seq...	4/5	80.0%	0

Verdict matrix

cvr_01_demo_structure	transcript_missing
cvr_02_assembly_disclosure	report_structure_complete
cvr_03_disclaimer_presence	report_structure_complete
cvr_04_limitations_section	report_structure_complete
cvr_05_evidence_trail	report_structure_complete

Persistent failures (1)

- cvr_01_demo_structure

Findings (1)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (1)

1/1 **cvr_01_demo_structure** transcript_missing

commit bb051158 · Transcript source missing · 1e19ecd31025

> Report does not cite a transcript (NM_*, ENST_*, or MANE Select). PVS1 LoF assessment is transcript-dependent.

FINDING CVR Phase 1 baseline — demo mode must produce a structurally complete report

HAZARD_METRIC An ACMG classification report missing any of reference build, transcript, data source versions, limitations, disclaimer, evidence trail, or disease context is unreviewable by a clinical auditor and must not be credited as a passing run.

FINDING_CATEGORY report_structure_complete

```
exit_code = 0
report_exists = True
assembly_mentions = ["GRCh38"]
transcript_mentions = []
clinvar_version = False
gnomad_version = True
disclaimer_present = True
limitations_present = True
… (4 more omitted)
```

RICHARDS_2015: Richards, S. et al. (2015). Standards and guidelines for the interpretation of sequence variants: a jo…

Chain of custody

Every verdict written by clawbio-bench carries an embedded SHA-256 self-hash. The sidecar `verdict_hashes.json` under each harness directory indexes those hashes independently, and `clawbio-bench --verify <results/>` runs a three-layer check (self-hash + sidecar + stdout/stderr log reconciliation).

Harness	Verdict hashes	Rubric size	Core ver	Run timestamp
bio-orchestrator	54	7	0.1.1	2026-04-06 04:09:13
equity-scorer	15	10	0.1.1	2026-04-06 04:09:29
nutrigx-advisor	10	9	0.1.1	2026-04-06 04:09:36
pharmgx-reporter	33	7	0.1.1	2026-04-06 04:09:39
claw-metagenomics	7	7	0.1.1	2026-04-06 04:09:48
clawbio-finemapping	16	16	0.1.1	2026-04-06 04:09:49
clinical-variant-reporter	5	10	0.1.1	2026-04-06 04:09:53

Total verdict hashes across all harnesses: **140**.

Environment fingerprint

Python	3.14.3
Platform	macOS-26.3.1-arm64-arm-64bit-Mach-O
Hostname hash	ae468cd6b8a7
Installed packages	34
Package-set SHA-256	77e1915e0950439029eb3a41
Audit target SHA	bb05115
Suite wall time	40.1 s

Appendix

Category counts

bio-orchestrator

stub_silent	3
stub_warned	9
unroutable_crash	1
unroutable_handled	13

equity-scorer

edge_crash	12
edge_handled	1

nutrigx-advisor

score_correct	4
snp_valid	3
threshold_consistent	2

pharmgx-reporter

correct_indeterminate	1
disclosure_failure	8
incorrect_determinate	1
incorrect_indeterminate	3
omission	2

claw-metagenomics

exit_handled	1
injection_blocked	4

clawbio-finemapping

credset_pip_is_alpha_mismatch	1
credset_purity_none_wrongly_pure	1
edge_handled	1
finemap_correct	5

input_validation_missing	2
pip_nan_silent	1
susie_moment_field_mislabeled	1
susie_nonconvergence_suppressed	1
susie_null_forced_signal	1
susie_spurious_secondary_signal	1

clinical-variant-reporter

transcript_missing	1

Test case inventory

bio-orchestrator (54 cases)

- raw("comp_01_pharmgx_nutrigx") · directory · 1 file
- raw("comp_02_equity_profile_report") · directory · 1 file
- raw("err_01_unknown_ext") · directory · 2 files
- raw("err_02_empty_query") · directory · 1 file
- raw("err_03_missing_file") · directory · 1 file
- raw("err_04_path_traversal_skill") · directory · 1 file
- raw("err_05_no_input") · directory · 1 file
- raw("err_06_nonexistent_skill") · directory · 1 file
- raw("err_07_dot_ext") · directory · 2 files
- raw("err_08_list_skills") · directory · 1 file
- raw("err_09_backslash_traversal") · directory · 1 file
- raw("err_10_ambiguous_query") · directory · 1 file
- raw("err_11_txt_unreachable") · directory · 2 files
- raw("ext_01_vcf") · directory · 2 files
- raw("ext_02_vcf_gz") · directory · 2 files
- raw("ext_03_csv") · directory · 2 files
- raw("ext_04_tsv") · directory · 2 files
- raw("ext_05_fastq") · directory · 2 files
- raw("ext_06_fastq_gz") · directory · 2 files
- raw("ext_07_fq") · directory · 2 files
- raw("ext_08_bam") · directory · 2 files
- raw("ext_09_pdb") · directory · 2 files
- raw("ext_10_cif") · directory · 2 files
- raw("ext_11_h5ad") · directory · 2 files
- raw("ext_12_png") · directory · 2 files
- raw("ext_13_cram") · directory · 2 files
- raw("ext_14_jpg") · directory · 2 files
- raw("ext_15_tiff") · directory · 2 files
- raw("force_01_clinical_variant_reporter") · directory · 1 file
- raw("force_02_variant_annotation") · directory · 1 file
- raw("force_03_clinical_trial_finder") · directory · 1 file
- raw("force_04_target_validation_scorer") · directory · 1 file
- raw("force_05_methylation_clock") · directory · 1 file
- raw("inj_01_routing_hijack") · directory · 1 file
- raw("inj_02_exfil_attempt") · directory · 1 file
- raw("inj_03_flock_routing_hijack") · directory · 1 file
- raw("kw_01_diversity") · directory · 1 file

- raw("kw_02_heim") · directory · 1 file
- raw("kw_03_fst") · directory · 1 file
- raw("kw_04_annotate_variant") · directory · 1 file
- raw("kw_05_alphafold") · directory · 1 file
- raw("kw_06_pubmed") · directory · 1 file
- raw("kw_07_single_cell") · directory · 1 file
- raw("kw_08_nextflow") · directory · 1 file
- raw("kw_09_heterozygosity") · directory · 1 file
- raw("kw_10_polygenic_risk") · directory · 1 file
- raw("kw_11_gwas_lookup") · directory · 1 file
- raw("kw_12_genome_compare") · directory · 1 file
- raw("kw_13_clinpgx") · directory · 1 file
- raw("kw_14_keyword_collision_variant") · directory · 1 file
- raw("kw_15_profile_report") · directory · 1 file
- raw("kw_16_pharmacogenomics_unreachable") · directory · 1 file
- raw("kw_17_nutrition_unreachable") · directory · 1 file
- raw("kw_18_metagenomics_unreachable") · directory · 1 file

equity-scorer (15 cases)

- raw("eq_01_fst_known_af") · directory · 3 files
- raw("eq_02_fst_identical") · directory · 3 files
- raw("eq_03_fst_monomorphic") · directory · 3 files
- raw("eq_04_fst_single_sample") · directory · 3 files
- raw("eq_05_het_all_het") · directory · 3 files
- raw("eq_06_het_all_hom") · directory · 3 files
- raw("eq_07_heim_balanced") · directory · 3 files
- raw("eq_08_heim_single_pop") · directory · 3 files
- raw("eq_09_heim_custom_weights") · directory · 3 files
- raw("eq_10_vcf_gz_crash") · directory · 3 files
- raw("eq_11_multiallelic") · directory · 3 files
- raw("eq_12_haploid_x") · directory · 3 files
- raw("eq_13_csv_ancestry") · directory · 2 files
- raw("eq_14_csv_unknown_pops") · directory · 2 files
- raw("eq_15_three_populations") · directory · 3 files

nutrigx-advisor (10 cases)

- raw("ng_01_all_risk_elevated") · directory · 2 files
- raw("ng_02_all_het_moderate") · directory · 2 files
- raw("ng_03_folate_elevated") · directory · 2 files
- raw("ng_04_caffeine_moderate") · directory · 2 files
- raw("ng_05_vitamin_d_elevated") · directory · 2 files
- raw("ng_06_repro_bundle") · directory · 2 files
- raw("ng_07_empty_input") · directory · 2 files
- raw("ng_08_no_panel_snps") · directory · 2 files
- raw("ng_09_allele_mismatch_bug") · directory · 2 files
- raw("ng_10_low_coverage") · directory · 2 files

pharmgx-reporter (33 cases)

- raw("cyp2b6_star18_hom") · file
- raw("cyp2c19_pm_clopidogrel") · file
- raw("cyp2c19_rapid_clopidogrel") · file
- raw("cyp2d6_del") · file
- raw("cyp2d6_duplication_xn") · file

- raw(“cyp2d6_normal”) · file
- raw(“cyp2d6_phase_ambiguous”) · file
- raw(“cyp2d6_pm”) · file
- raw(“cyp2d6_star13_hybrid”) · file
- raw(“cyp2d6_star5_deletion”) · file
- raw(“cyp3a5_7_ins”) · file
- raw(“dpyd_2a_het”) · file
- raw(“dpyd_2a_hom”) · file
- raw(“dpyd_absent”) · file
- raw(“dpyd_neg”) · file
- raw(“dpyd_partial”) · file
- raw(“empty_no_pgx”) · file
- raw(“g6pd_a_minus_indeterminate”) · file
- raw(“grch37_reference_mismatch”) · file
- raw(“hla_b5701_indeterminate”) · file
- raw(“inj_cyp2c19_header_tampering”) · file
- raw(“mt_rnr1_1555g_indeterminate”) · file
- raw(“nudt15_star3_hom”) · file
- raw(“phi_patient_identifiers_in_header”) · file
- raw(“slco1b1_statin_myopathy”) · file
- raw(“tpmt_compound_het”) · file
- raw(“ugt1a1_28_absent”) · file
- raw(“ugt1a1_28_het”) · file
- raw(“ugt1a1_28_hom”) · file
- raw(“ugt1a1_neg”) · file
- raw(“warfarin_missing_both”) · file
- raw(“warfarin_missing_vkorc1”) · file
- raw(“warfarin_normal”) · file

claw-metagenomics (7 cases)

- raw(“mg_01_demo_functional”) · directory · 1 file
- raw(“mg_02_no_shell_true”) · directory · 1 file
- raw(“mg_03_commands_sh_safety”) · directory · 1 file
- raw(“mg_04_env_var_injection”) · directory · 1 file
- raw(“mg_05_exit_suppressed”) · directory · 1 file
- raw(“mg_06_adversarial_path”) · directory · 1 file
- raw(“mg_07_dash_prefix”) · directory · 1 file

clawbio-finemapping (16 cases)

- raw(“fm_01_abf_single_causal”) · directory · 2 files
- raw(“fm_02_susie_single_causal”) · directory · 2 files
- raw(“fm_03_susie_null_locus”) · directory · 2 files
- raw(“fm_04_susie_phantom_secondary”) · directory · 2 files
- raw(“fm_05_abf_variant_n_collapsed”) · directory · 2 files
- raw(“fm_06_abf_nan_zero_se”) · directory · 2 files
- raw(“fm_07_abf_nonpositive_n”) · directory · 2 files
- raw(“fm_08_susie_negative_n”) · directory · 2 files
- raw(“fm_09_credset_pip_mismatch”) · directory · 2 files
- raw(“fm_10_credset_purity_mean_hides_weak”) · directory · 2 files
- raw(“fm_11_credset_purity_none_wrongly_pure”) · directory · 2 files
- raw(“fm_12_susie_nonconvergence”) · directory · 2 files
- raw(“fm_13_credset_coverage_zero”) · directory · 2 files
- raw(“fm_14_susie_moment_mislabeled”) · directory · 2 files
- raw(“fm_15_abf_extreme_z”) · directory · 2 files

- raw("fm_16_credset_coverage_field") · directory · 2 files

clinical-variant-reporter (5 cases)

- raw("cvr_01_demo_structure") · directory · 1 file
- raw("cvr_02_assembly_disclosure") · directory · 1 file
- raw("cvr_03_disclaimer_presence") · directory · 1 file
- raw("cvr_04_limitations_section") · directory · 1 file
- raw("cvr_05_evidence_trail") · directory · 1 file

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